SOC-5811 Week 5: Sampling distributions and statistical inference

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Review

Why are we creating models (e.g., linear regression) for data at all?

- Summarizing relationships.
- Making predictions.

Review

1. Choose a functional form for the model.

$$pcthouse_i = \beta_0 + \beta_1 pctpop_i + \epsilon_i$$

Once we define a model, how do we estimate the parameters using data?

Review

1. Choose a functional form for the model.

$$pcthouse_i = \beta_0 + \beta_1 pctpop_i + \epsilon_i$$

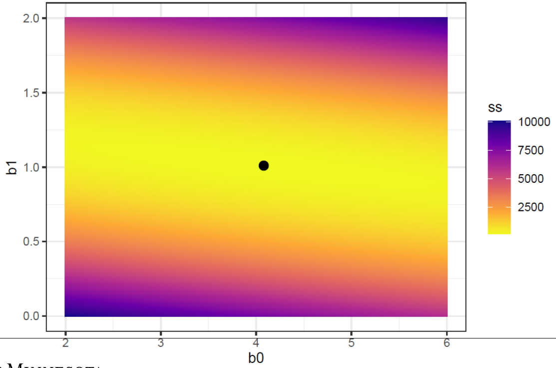
2. Choose an estimator (i.e., objective function) that relates the model to real observed data (e.g., OLS).

$$\text{Sum of squared residuals} = \sum_{i=1}^{n} (y_i - \widehat{\beta_0} - \widehat{\beta_1} x_i)^2$$

3. Fit the model (e.g., estimate parameters) by optimizing the objective function.

What is happening behind the scenes to fit the model?

- If there is a closed-form solution: math.
- If there is not a closed-form solution: iterative calculations and approximations.



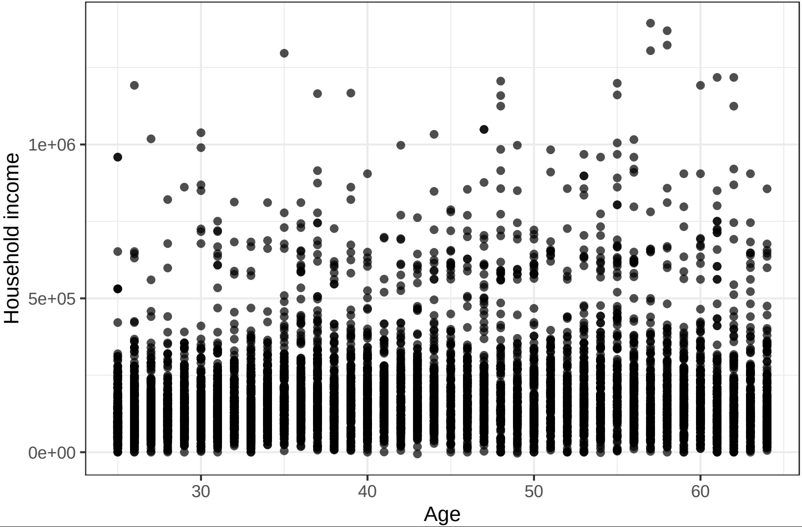
Review

- Linear regression is practically very useful because it summarizes relationships with simple parameters that are easy to interpret.
- This is true even once we start adding *more* independent variables!

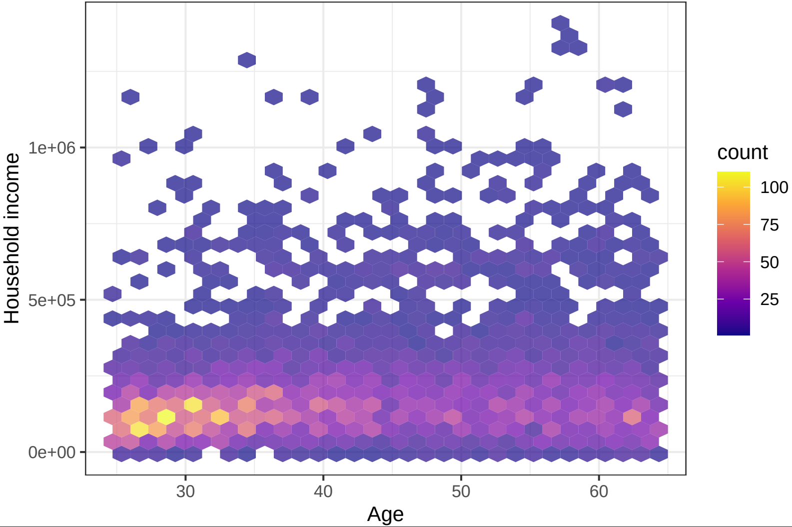
Statistical inference

- Across working years, are incomes higher at older ages in Hennepin County?
- If we can agree on definitions of income and age, then this seems like a question we should be able to answer empirically!

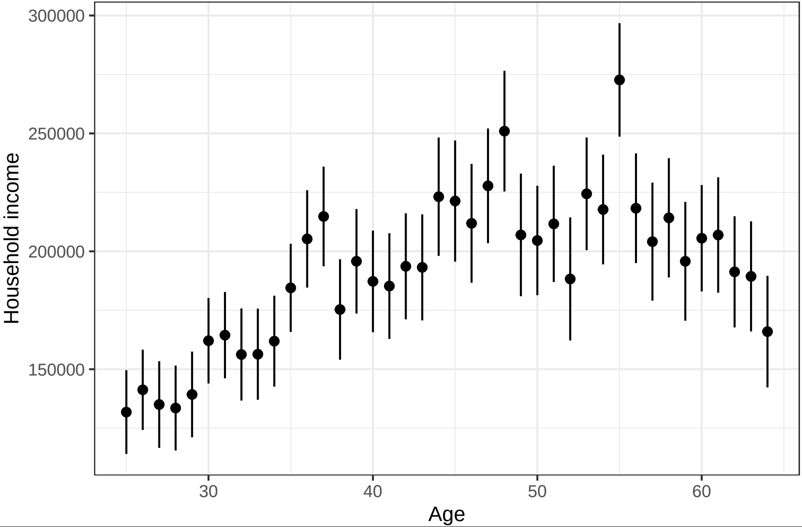
STATISTICAL INFERENCE



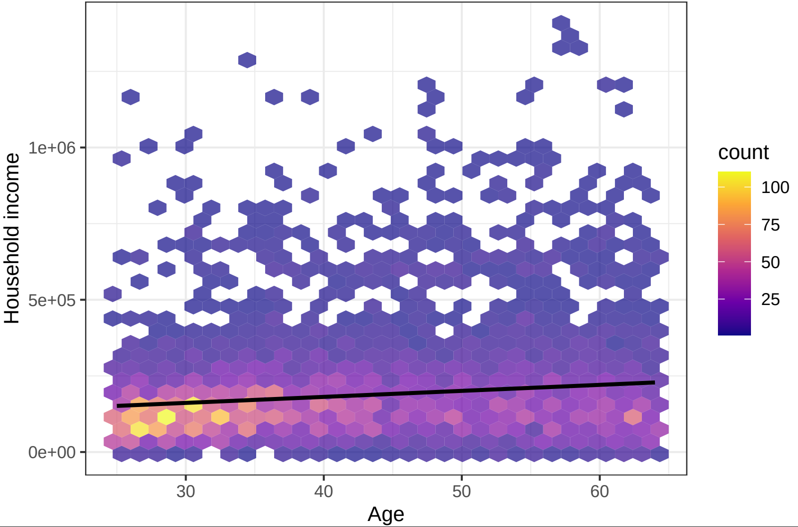
Statistical inference



Statistical inference



Statistical inference



Statistical inference

- Could the sample relationship be due to chance?

Statistical inference

- Could the sample relationship be due to chance?
- Based on our *sample*, how sure are we that the *population* parameter is different from 0?

- **Parameters** = the unknown numbers that determine a statistical model.

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- Parameters can be used to simulate new data from the model.

Statistical inference

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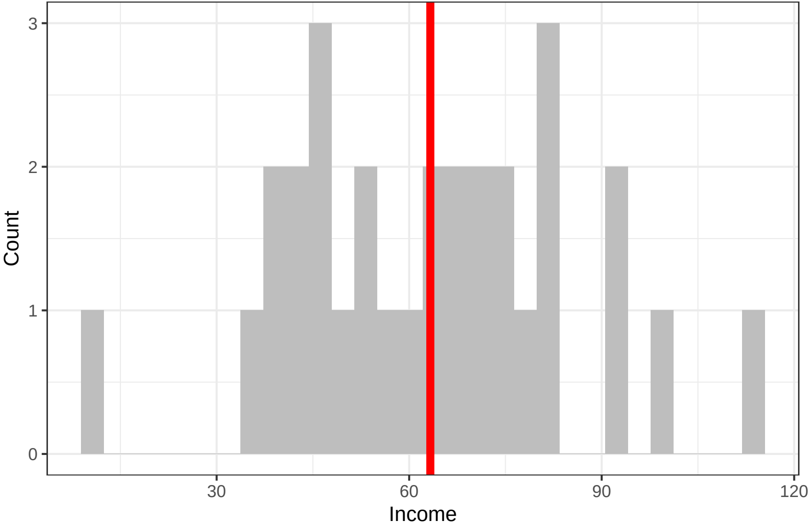
Statistical inference

- **Statistical inference** = a set of operations on data that yield estimates and uncertainty statements about **predictions and parameters** of some underlying process or population.
- More simply, we observe data and we try to learn something about where it came from (i.e., the data generating process).
- **This involves uncertainty.**

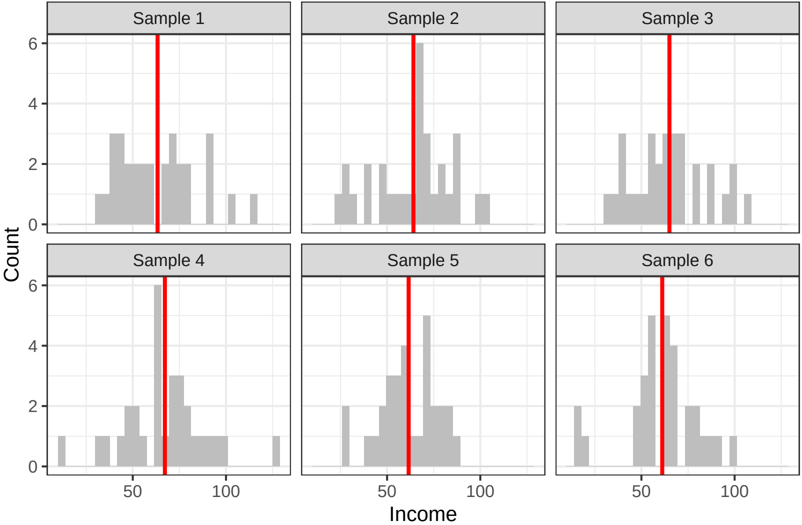
Statistical inference

- Say I have a population of 1000 people and I ask 30 of them their income. The parameter I'm interested in inferring something about is the average income in the entire population.

Statistical inference



Statistical inference



Sampling distribution

- **Sampling distribution** = the set of possible datasets that could have been observed if the data collection process had been re-done, along with the probabilities of these values.

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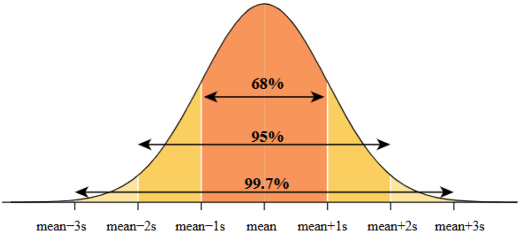
- In practice, we will not know the sampling distribution; we can only estimate it.

Standard errors

- **Standard deviation** = a measure of the spread of data around the mean.

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$



Standard errors

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$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

- **Standard error** = ???

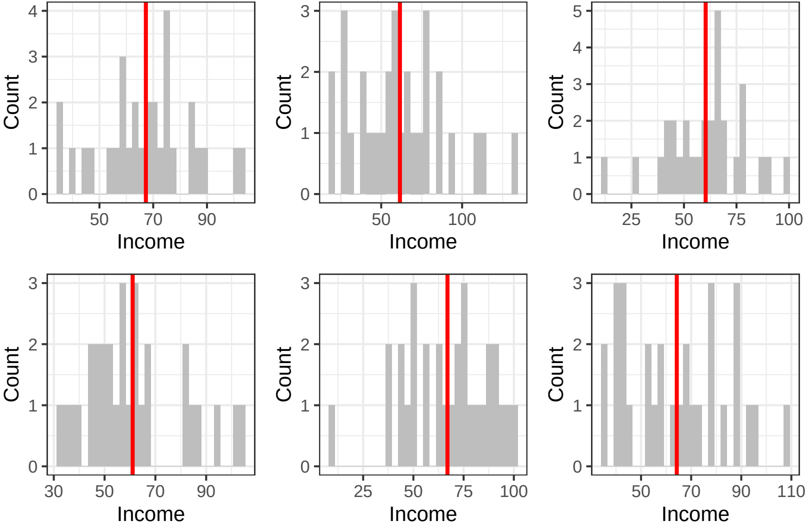
STANDARD ERRORS

- **Standard deviation** = a measure of the spread of data around the sample mean.

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

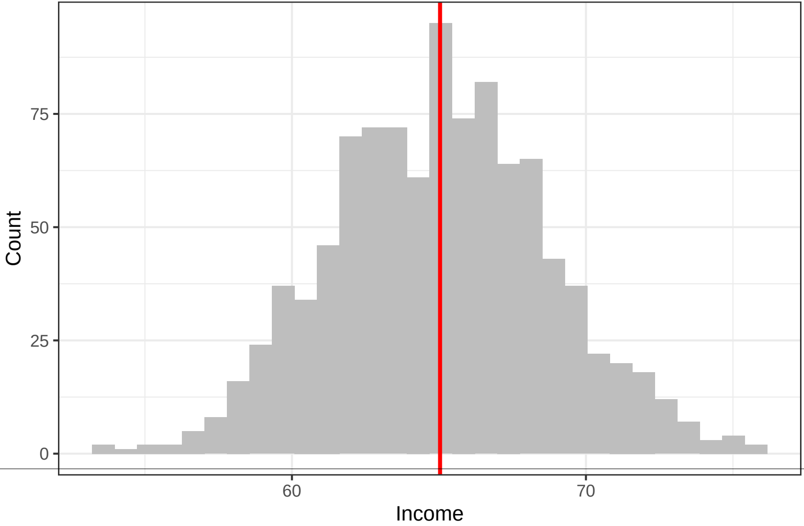
- **Standard error** = a measure of the spread of sample means around the population mean; the standard deviation of the sampling distribution.

Sampling distribution



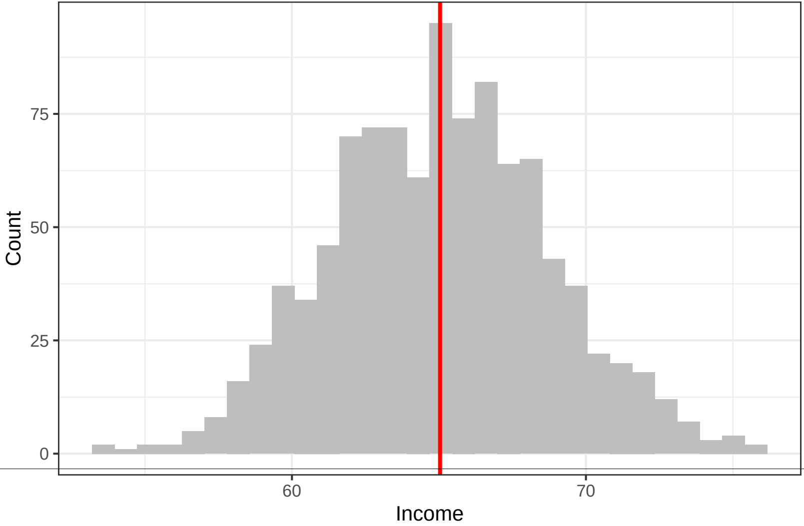
Sampling distribution

Sampling distribution of the mean:

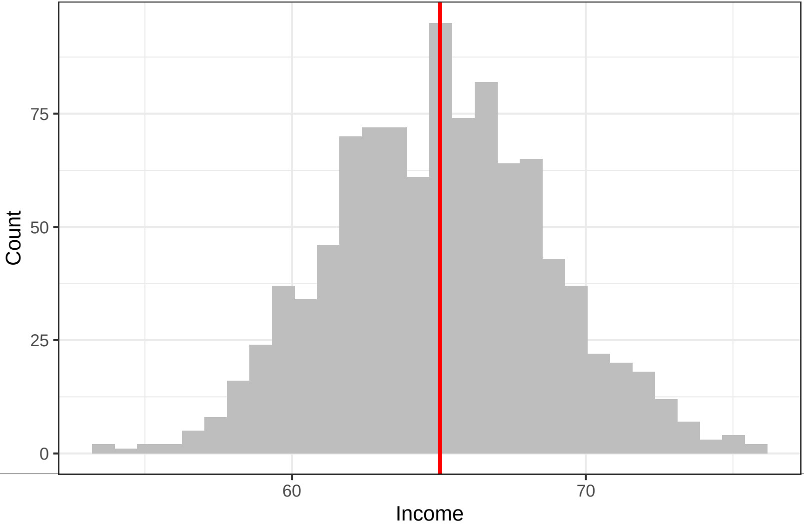


Sampling distribution

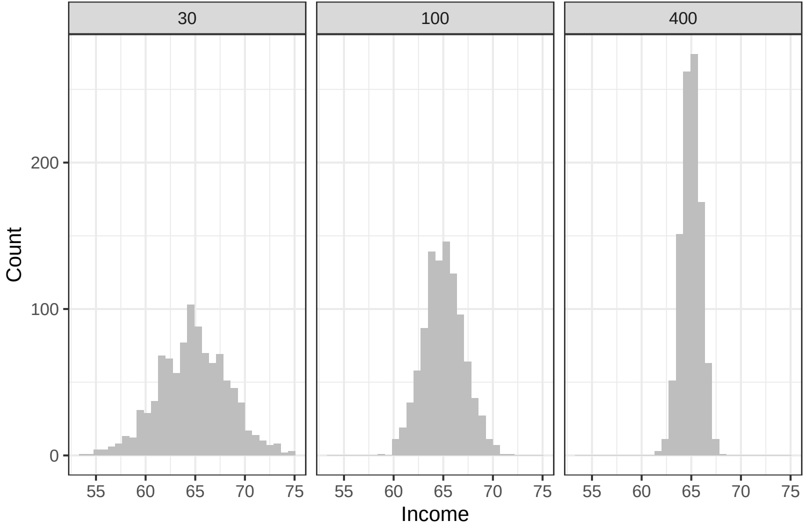
What do you notice about the shape of this distribution?



- **Central Limit Theorem**

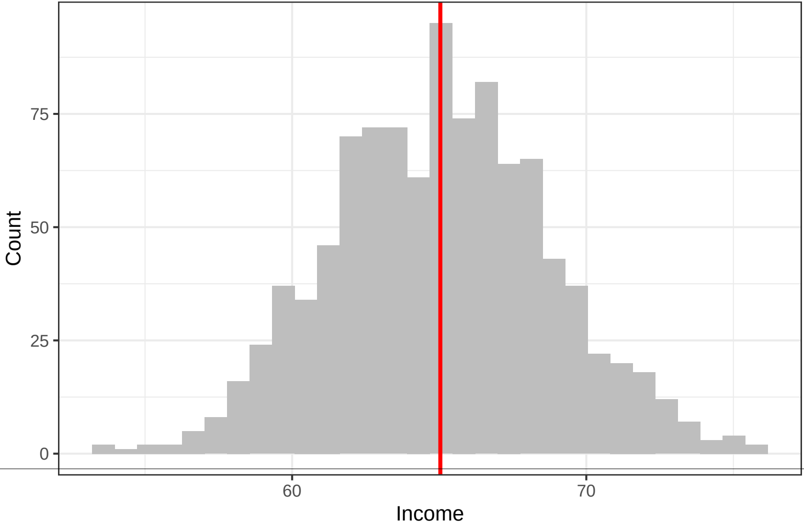


Sampling distribution



SAMPLING DISTRIBUTION

How do I interpret the standard deviation of this distribution?



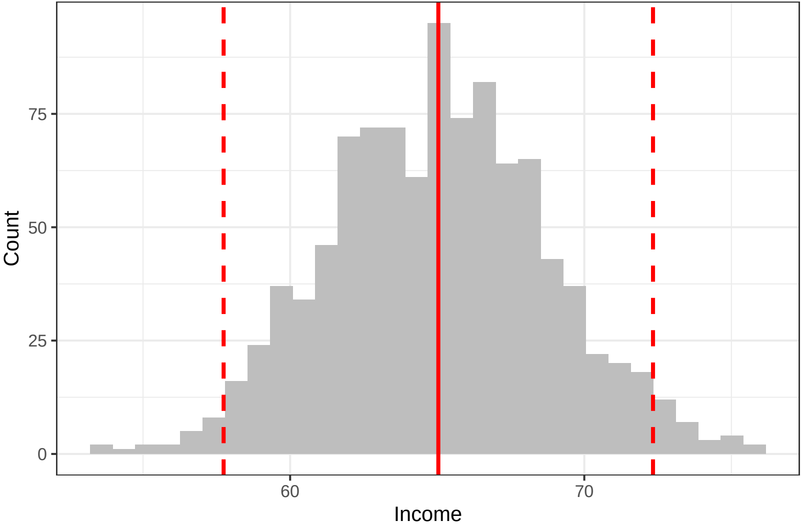
Sampling distribution

- However defined, the standard error is a measure of the variation in an estimate and gets smaller as sample size gets larger, converging on zero as the sample increases in size.

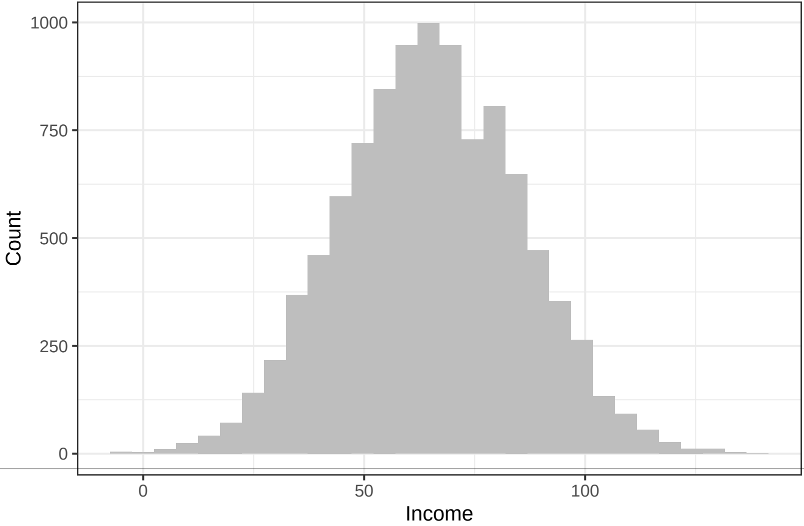
Sampling distribution

- However defined, the standard error is a measure of the variation in an estimate and gets smaller as sample size gets larger, converging on zero as the sample increases in size.
- The **confidence interval** represents a range of values of a parameter or quantity of interest that are roughly consistent with the data, given the assumed sampling distribution.

Sampling distribution

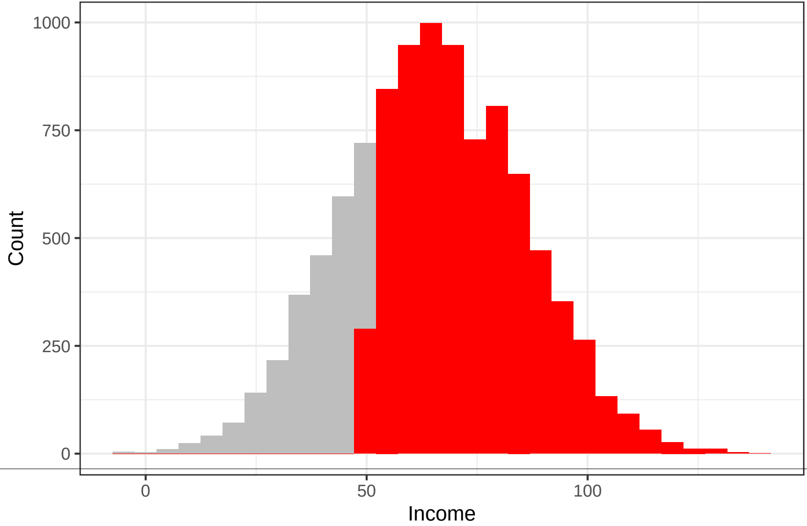


It is possible to be very confidently wrong!



Bias

Say only people with incomes above 50k complete my survey:



Describing uncertainty:

- Standard errors capture a particular (but very important!) type of uncertainty: sampling error.

Bias

Describing uncertainty:

- Standard errors capture a particular (but very important!) type of uncertainty: sampling error.
- There are lots of other reasons to be uncertain!

What can we do with standard errors?

- Hypothesis testing

Review

- Standard error

Review

- Standard error
  - The standard deviation of an estimate

Review

- Standard error
  - The standard deviation of an estimate
  - How close are our sample statistics to the true statistics?

Review

- Standard error
  - The standard deviation of an estimate
  - How close are our sample statistics to the true statistics?
  - Gets smaller as sample size gets larger, converging on zero as sample size increases.

Review

- Confidence intervals

Review

- Confidence intervals
  - Mean +/- 2 standard errors

Review

- Confidence intervals
  - Mean +/- 2 standard errors
  - A 95% confidence interval has the following property: if we take repeated samples and construct the confidence interval for each sample, 95% of the intervals will contain the true unknown value of the parameter.

Review

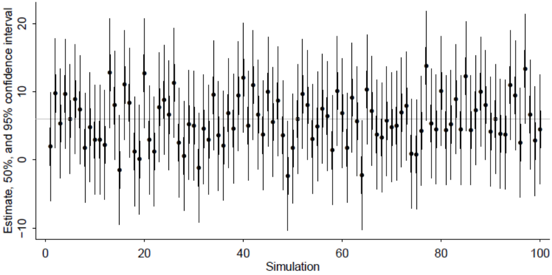


Figure 4.2 *Simulation of coverage of confidence intervals: the horizontal line shows the true parameter value, and dots and vertical lines show estimates and confidence intervals obtained from 100 random simulations from the sampling distribution. If the model is correct, 50% of the 50% intervals and 95% of the 95% intervals should contain the true parameter value, in the long run.*

$$y_i = \beta_0 + \beta_1 X + \epsilon_i$$

$$\beta_1 = 5.5\,(95\%\,\text{CI: }4.0 - 7.0)$$

What is our model saying about changes in Y related to changes in X?

Predictions: If X goes up by 1, Y will increase somewhere between 4.0 and 7.0.

Hypothesis testing

$$y_i = \beta_0 + \beta_1 X + \epsilon_i$$

$$H_0 : \text{There is no relationship between X and Y}$$

$$H_1 : \text{There is some relationship between X and Y}$$

Hypothesis testing

$$y_i = \beta_0 + \beta_1 X + \epsilon_i$$

$$H_0 : B_1 = 0$$

$$H_1 : B_1 \neq 0$$

Hypothesis testing

- We need to determine whether our estimate of β_1 is sufficiently far from zero such that we can be confident that β_1 is non-zero.

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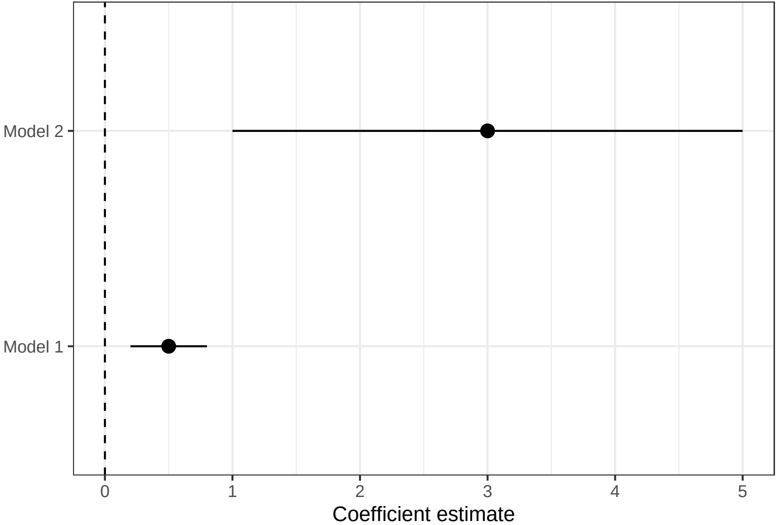
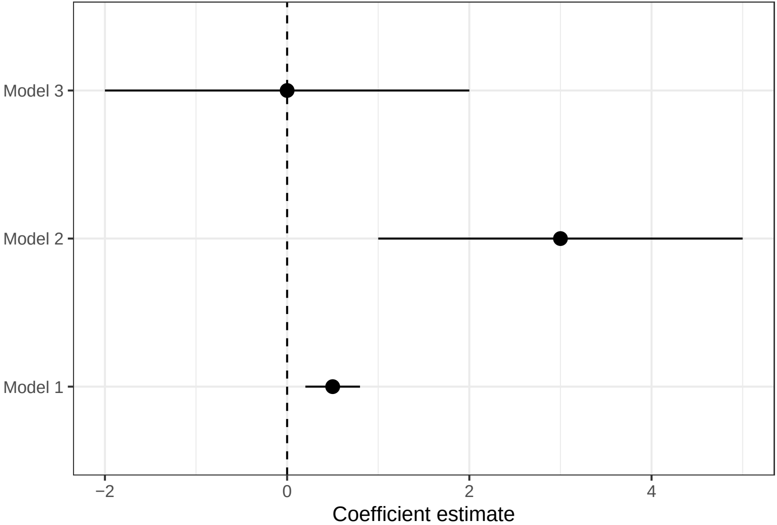
- We need to determine whether our estimate of β_1 is sufficiently far from zero such that we can be confident that β_1 is non-zero.
- **How far is far enough?** It depends on the accuracy of our estimate of β_1; that is, it depends on the standard error of β_1.

Hypothesis testing



Hypothesis testing



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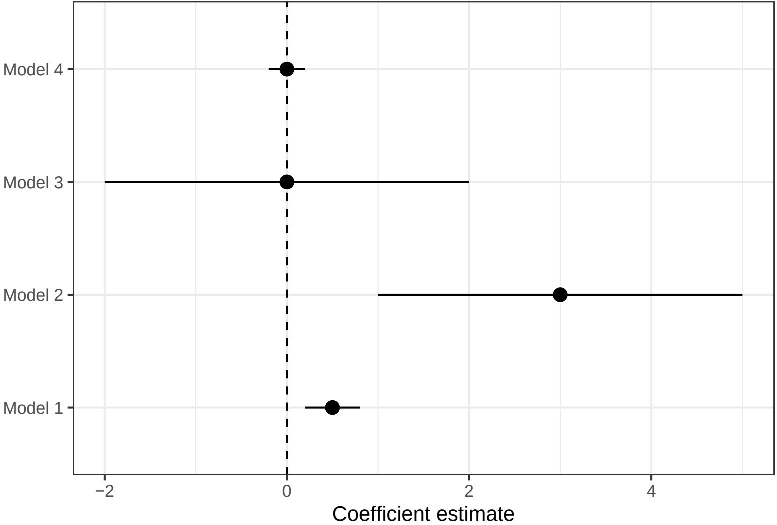

P-VALUES

- Is it "statistically significant?"

P-VALUES

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- If $\beta_1 = 0$, how likely is it that we would estimate a β_1 as large or larger than the one we did observe?

P-VALUES

- **p-value** = the probability that the null model would produce a parameter estimate as extreme as your observed estimate.

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- If the null model is true and we repeated our data collection many times, how often would we observe a parameter estimate as extreme as our observed estimate by chance?

P-VALUES

- **p-value** = the probability that the null model would produce a parameter estimate as extreme as your observed estimate.
- If the null model is true and we repeated our data collection many times, how often would we observe a parameter estimate as extreme as our observed estimate by chance?
- A small p-value indicates that it is unlikely to observe such a substantial association between the predictor and the response due to chance, in the absence of any real association between the predictor and the response. Hence, if we see a small p-value, then we can infer that there is an association between the predictor and the response.

P-VALUES

- *Statistical significance* is conventionally defined as a p-value less than 0.05, relative to some null hypothesis or prespecified value that would indicate no effect present.

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- *Statistical significance* is conventionally defined as a p-value less than 0.05, relative to some null hypothesis or prespecified value that would indicate no effect present.
- An estimate is said to be *not* statistically significant if the observed value could reasonably be explained by simple chance variation.
- The possible outcomes of a hypothesis test are "reject" or "not reject." It is never possible to "accept" a statistical hypothesis, only to find that the data are not sufficient to reject it.

Imagine flipping a coin 100 times and interpreting the results relative to some null probability model.

What would the null model be in this context?

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Null model: $P(\text{heads}) = 0.5$; this is a fair coin.

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Results: 60 heads, 40 tails.

P-VALUES

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Null model: $P(\text{heads}) = 0.5$; this is a fair coin.

Results: 55 heads, 45 tails.

Would my p-value be large or small?

Imagine flipping a coin 100 times and interpreting the results relative to some null probability model.

Null model: $P(\text{heads}) = 0.5$; this is a fair coin.

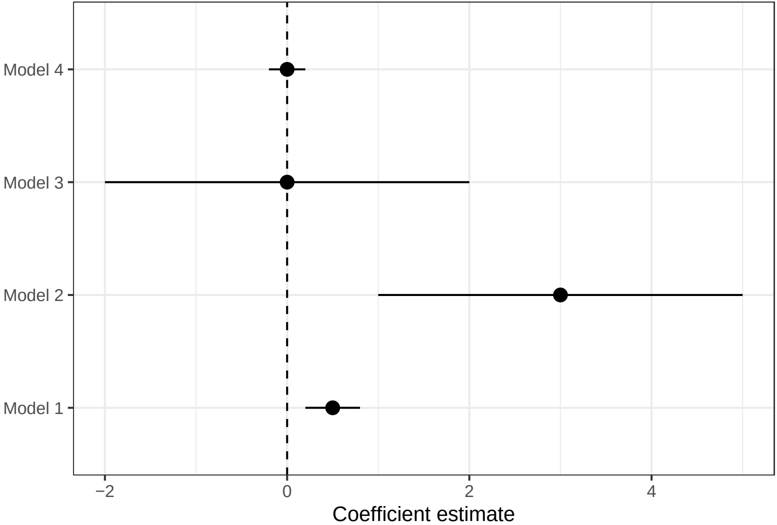
Results: 70 heads, 30 tails.

Would my p-value be large or small?

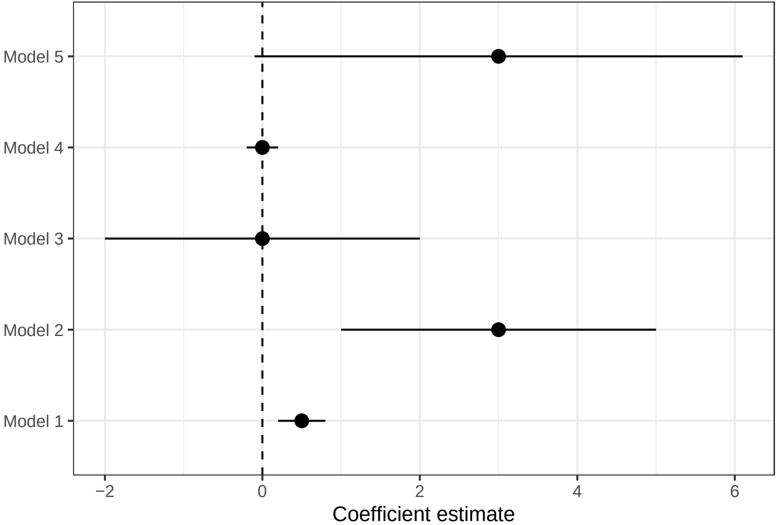
P-VALUES

- Does a small p-value mean that the observed association between X and Y is *important*?
- Which of these models provides “better evidence” that X is associated with Y?

P-VALUES

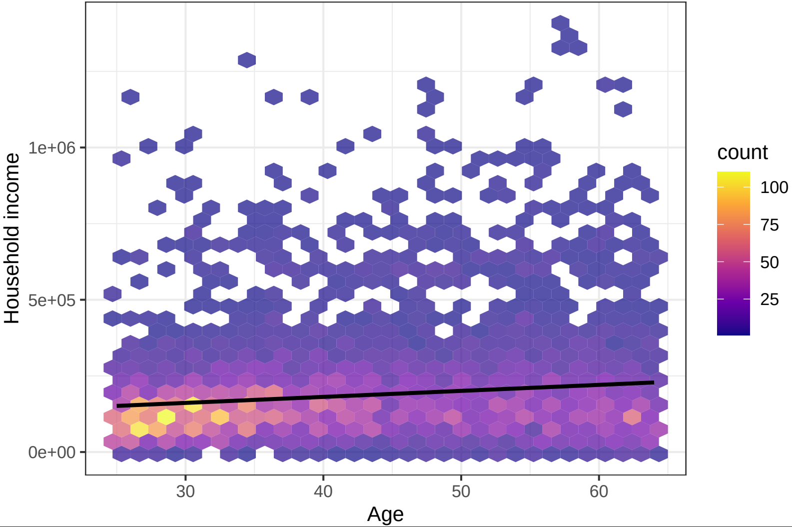


P-VALUES



Model 5
Model 4
Model 3
Model 2
Model 1
−2
0
2
4
6
Coefficient estimate

P-VALUES



Household income
1e+06
5e+05
0e+00
30
40
50
60
Age
count
100
75
50
25

P-VALUES

```
##
## Call:
## lm(formula = hhincome ~ age, data = msp_micro[educ == "4 years of college" &
##     age %in% 25:64 & !is.na(hhincome), ])
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -226527  -86525  -31445   37294 1179553
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 102162.8     6482.6   15.76   <2e-16 ***
## age           1974.0      148.9   13.26   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 149700 on 7161 degrees of freedom
## Multiple R-squared:  0.02397,    Adjusted R-squared:  0.02383
## F-statistic: 175.8 on 1 and 7161 DF,  p-value: < 2.2e-16
```

P-VALUES

$$\beta_1 = 5.5\,(95\%\,\text{CI: } 4.0 - 7.0)^*$$

$$\beta_1 = 5.5\,(0.75)^*$$

P-VALUES

Table 2. Children with Same-Sex Parents and Standardized Test Scores at the End of Primary Education

	All Children		Children Raised by Same-Sex Parents from Birth[a]		
	(1)	(2)	(3)	(4)	(5)
Child has same-sex parents (1 is yes)	.106***	.054**	.194***	.139***	.147***
	(.019)	(.018)	(.024)	(.023)	(.041)
Sex (1 is male)	−.004*	−.009***	−.003	−.007***	
	(.002)	(.002)	(.002)	(.002)	
Ethnicity (ref: both parents born in NL)					
One parent NL, other Western country		.004		.004	
		(.003)		(.003)	
One parent NL, other non-Western country		−.031***		−.031***	
		(.005)		(.005)	
Both parents not NL		−.111***		−.111***	
		(.006)		(.006)	
Parental education (1 is diploma SE)		.531***		.532***	
		(.004)		(.004)	
Log household income		.083***		.084***	
		(.002)		(.002)	
Mean parental age		.018***		.018***	
		(.000)		(.000)	
Family size		−.066***		−.066***	
		(.001)		(.001)	
Family structure (ref: married parents)					
Cohabiting parents		−.073***		−.073***	
		(.002)		(.002)	
Other		−.367***		−.154***	
		(.012)		(.032)	
Fixed effects					
Birth year	Yes	Yes	Yes	Yes	
Neighborhood	No	Yes	No	Yes	
Method	OLS	OLS	OLS	OLS	CEM[b]
Number of children	1,204,692	1,204,692	1,195,624	1,195,624	48,388
Number of children with same-sex parents	2,971	2,971	1,390	1,390	1,390
R^2	.004	.108	.004	.109	.121

- Publication norms

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